



RAMCO INSTITUTE OF TECHNOLOGY

(Approved by AICTE - New Delhi and Affiliated to Anna University - Chennai)
NAAC Accredited and An ISO 9001:2015 certified institution



RIT/MECH/VAC/2021/03

27.07.2021

To

The Director,

Centre for Academic Courses,

Anna University,

Chennai - 600025.

Respected Sir,

Sub: Requesting permission to offer below mentioned approved Value Added Course-
V Semester B.E. Mechanical Engineering – Academic Year 2021-2022 – reg.

Ref: List of approved Value Added Courses – Letter No: 2525/AU/VA/CAC/2019 on
01.07.2019.

With reference to the above, we have proposed to offer the following approved value added courses to our V semester B.E Mechanical Engineering Students for the academic year 2021 – 2022 (ODD Semester). The syllabus and schedule of the approved value added courses along with approved letter received from Centre for Academic Courses, Anna University, Chennai are enclosed for your kind perusal.

Sl. No.	Course Code	Course Name	Credits
1	MVA006	GREEN ENERGY TECHNOLOGIES AND MANAGEMENT	2
2	MVA009	SURFACE COATING TECHNOLOGY	2

Hereby, we request your kind approval and permission to offer these value added courses to our V semester B.E Mechanical Engineering Students for the academic year 2021 – 2022 (ODD Semester).

Thanking you,

Yours faithfully,




HOD/Mech


Vice Principal


Principal



CENTRE FOR ACADEMIC COURSES
ANNA UNIVERSITY
CHENNAI - 600 025

Off: 22357077 / 73

22357074

Fax / Dir : 22352272

Dr. R. RAJU
DIRECTOR

Letter No 2525/AU/VA/CAC/2019

01.07.2019

To
The Controller of Examinations
Anna University
Chennai - 25

Sir,

Sub : A U - CAC - Affiliated Institutions - Value Added Course - Reg.
Ref : Letter No.RIT/MECH/VAC/2019/02 dated:20.06.2019.

With reference to the letter cited, the following Value Added Course offered by Ramco Institute of Technology, Rajapalayam, Affiliated Institutions is allotted the course code as detailed below.

SL.No	Code Allotted	Title
1	MVA005	Smart Materials and Structures
2	MVA006	Green Energy Technologies and Management
3	MVA007	Automation Suite for Smart Systems
4	MVA008	Internet of Things Applications in Mechanical Engineering
5	MVA009	Surface Coating Technology

This is for your kind information and necessary action at your end.

Yours faithfully,

DIRECTOR

Copy to:

1. The Chairperson, Faculty of Mechanical Engineering, A.U., Ch -25.
2. The Principal, Ramco Institute of Technology, North Venganallur, Rajapalayam - 626 117.
3. The Stock File - CAC.



CENTRE FOR ACADEMIC COURSES

ANNA UNIVERSITY
CHENNAI - 600 025

Off: 22357077 / 73

22357074

Fax / Dir : 22352272



Dr. S. HOSIMIN THILAGAR
DIRECTOR
Letter No.2649/AU/VA/CAC/2021

PME

6921



02.09.2021

Copy to - Ex cell
Acc. Coordinator

To
The Controller of Examinations
Anna University
Chennai - 25.

Sir/Madam,

Sub: A.U. - CAC - Affiliated Institutions - Value Added Course - Reg.

Ref: Letter No.RIT/MECH/VAC/2021/04 dated 27.07.2021.

With reference to the letter cited, the following Value Added Course offered by Ramco Institute of Technology, Virudhunagar District, Affiliated Institution is allotted the Course Code as detailed below.

SL. No.	Code Allotted	Title	Credits			
			L	T	P	C
1.	MVA023	IoT and Augmented Reality Applications in Mechanical Engineering	1	0	2	2

This is for your kind information and necessary action at your end.

Yours faithfully,

[Signature]
2/9/2021

DIRECTOR

Copy to:

1. The Chairperson, Faculty of Mechanical Engineering, A.U., Ch -25.
2. The Principal, Ramco Institute of Technology, North Venganallur, Ayyanarkovil Road, Rajapalayam - 626 117.
3. The Stock File - CAC.

[Signature]
2/9/2021

OBJECTIVES:

- To understand advanced and emerging technologies.
- To know about the key components that makes up an IoT system.
- To gain the knowledge and skills acquired to build and test a IoT system involving prototyping, programming and data analysis.
- To understand where the IoT & AR concept fits within the broader mechanical Engineering domain and possible future trends.

I FUNDAMENTALS AND APPLICATIONS OF IoT**10**

Introduction to IoT technology-Definition, Characteristics, Physical design, Logical design, Functional blocks, Communication models & APIs-IoT - Machine to Machine, Difference between IoT and M2M, Software defined Network - Network & Communication aspects - Home automation, Industry applications, Surveillance applications, Other IoT applications – Case Studies- Simulation of basic electronic components.

II ELEMENTS OF IoT**10**

Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensors and Transducers, Actuation, I/O interfaces. Software Components- Programming API's (using Python/Node.js/Arduino) for Communication. Protocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP- Python programming for Raspberry Pi.

III THE WORLD OF THE FUTURE – IoT & AUGMENTED REALITY**10**

Humanistic Intelligence-Wearable Computing and IoT (Internet of Things) - Overview of Mobile and Wearable Computing, Augmented Reality, and Internet of Things. The fundamental axes of the Wearables + IoT + AR space - Free-roaming AR: Wearable Computing, Wireless, Sensing, and Metasensing with light bulbs Phenomenal Augmented Reality. Fundamental basis of mobile and wearable AR. Internet of Everything – The Future and perspectives – Challenges and Usage of augmented / virtual reality tools.

TOTAL: 30 PERIODS**OUTCOMES:**

On successful completion of the course, the student will be able to:

- Describe the concepts of Internet of Things
- Analyze the basic protocols in wireless sensor network
- Design IoT applications in mechanical engineering domain and to analyze their performance
- Implement basic IoT and AR applications using hardware components.

Assessment Method (Weightage 100 %)			
In Semester	Assessment I	Assessment II	
	MCQ/Short Q & A	MCQ/Simple Experiments	Total
	50%	50%	100%

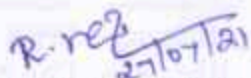
22/21

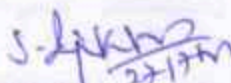
REFERENCES:

1. Waltenegus Dargie, Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice", Wiley Series on Wireless Communication and Mobile Computing, 2018.
2. Michael Margolis, Arduino Cookbook, Recipes to Begin, Expand, and Enhance Your Projects, second Edition, O'Reilly Media, 2011.
3. Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), Architecting the Internet of Things, Springer, 2011.
4. Tom Dieck, M. Claudia, Jung, Timothy, Augmented Reality and Virtual Reality, Springer, 2019.
5. Vijay Madiseti, Arshdeep Bahga, "Internet of Things: A Hands-On Approach" ISBN:0996025510, 2014.

WEB REFERENCE

1. <https://www.udemy.com/course/introduction-to-internet-of-things-iot-using-arduino/>
2. <https://www.edx.org/course/introduction-to-the-internet-of-things-iot?index=product&queryID=b01b6dd1a4e7ef3a89961070d1711c59&position=1>
3. <https://www.udemy.com/course/develop-augmented-reality-book-ar-business-card-with-unity/>


Course Designer & Coordinator


HOD/ Mechanical


Vice-Principal


Principal

RAMCO INSITIUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

VALUE ADDED COURSE TIME TABLE

Academic year 2021-22 (Odd Semester)

Title of the Course : **MEVAC1906 - IoT AND AR APPLICATIONS IN MECHANICAL ENGINEERING**

Year / Branch / Semester: **III / Mechanical / V**

Course Handling Faculty

Industrial Experts :

1. Mr.Moorthi Kanagaraj, Product Manager, M/s.Enthu Technology Solutions, Coimbatore.
2. Mr.K.S.Rathina Samy, Senior Technical Manager, M/s. Enthi Technology Solutions, Coimbatore.
3. Mr.S.Mohamed Safiyur Rahman, Software Project Manager, Robert Bosch Engineering and Business Solutions India (RBEI) Pvt. Ltd., Coimbatore.

Eminent Academicians from other Institutions :

1. Mr.K.Thirumoorthy, Assistant Professor (Sr. Grade), Mepco Schlenk Engineering College, Sivakasi
2. Mrs.S.Amutha, Assistant Professor(Sr. Grade), Mepco Schlenk Engineering College, Sivakasi

Internal Faculty Members :

1. Mr.Jerold John Britto, Assistant Professor(Sr. Grade)/ Ramco Institute of Technology
2. Mr.R.Venkatesh, Assistant Professor/ Ramco Institute of Technology
3. Mr.R.Prabhakaran, Assistant Professor/ Ramco Institute of Technology

Evaluation Team Members: 1. Head of the Department
2. One senior faculty member handling the course
3. One senior faculty member nominated by the Head of the Institution

Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
I	21.08.2021 (Saturday)	Introduction to IoT technology- Definition, Characteristics, Physical design, Logical design, Functional blocks, Communication models & APIs-IoT.	3	-
II	22.08.2021 (Sunday)	Machine to Machine, Difference between IoT and M2M, Software defined Network - Network & Communication aspects	3	-
III	28.08.2021 (Saturday)	Home automation, Industry applications, Surveillance applications, and Other IoT applications – Case Studies.	3	-
IV	29.08.2021 (Sunday)	Hands on Training - Simulation of basic electronic components using Thinker CAD,	-	3

02/22/21

Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
V	04.09.2021 (Saturday)	Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensors and Transducers, Actuation	3	-
VI	05.09.2021 (Sunday)	I/O interfaces. Software Components- Programming API's (using Python/Node.js/Arduino) for Communication.- Protocols- MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP	3	-
VII	11.09.2021 (Saturday)	Hands on Training - An online platform to bring Circuit assemblies to life virtually.	-	3
Assessment I	12.09.2021 (Sunday)	The assessment will be conducted as descriptive /MCQ type covers Session I to VII		
VIII	18.09.2021 (Saturday)	Hands on training – Python, programming for Raspberry Pi- LCD Integration with Raspberry Pi- Controlling LED with a Switch using Raspberry Pi	-	3
IX	19.09.2021 (Sunday)	Humanistic Intelligence-Wearable Computing and IoT (Internet of Things) - Overview of Mobile and Wearable Computing, Augmented Reality, and Internet of Things.	3	-
X	25.09.2021 (Saturday)	The fundamental axes of the Wearables + IoT + AR space - Free-roaming AR: Wearable Computing, Wireless, Sensing, and Metasensing with light bulbs Phenomenal Augmented Reality.	3	-
XI	26.09.2021. (Sunday)	Fundamental basis of mobile and wearable AR. Internet of Everything – The Future and perspectives - Challenges	3	-
XII	02.10.2021 (Saturday)	Hands on training – Usage of augmented / virtual reality tools	-	3
Assessment II	03.10.2021 (Sunday)	The assessment will be conducted as practical / descriptive/ MCQ type covers Session VIII to XII.		
Number of Periods for Theory and Practical Session (Practical session will be consider as 50% of the total periods)			24	12
Total number of periods (2 credit course)			30	

R.vel
27/07/21
Course Designer & Coordinator

S. Lakshmi
27/7/21
HoD/Mechanical

S. Lakshmi
27/7/21
Vice-Principal

M. Lakshmi
27/7/21
Principal



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF MECHANICAL ENGINEERING

Academic year 2021-22 (Odd Semester)

Anna University approved Value Added Course

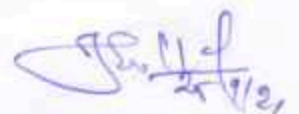
Willingness Details

Title of the Course : MVA023- IoT and AR Applications in Mechanical Engineering

Year / Branch / Semester: III / Mechanical / V

S. No.	Register Number	Name of the Students	Signature
1.	953619114003	G ARULKUMAR	G. Arulkumar
2.	953619114005	ARUN PRASATH T	A. Prasad
3.	953619114006	S.ATHISHANKAR	S. Athishankar
4.	953619114009	BHARANIDHARAN R	B. Bharanidharan
5.	953619114011	BHARATH M	B. Bharath
6.	953619114013	S S DEV PRASSATH	S. S. Dev Prassath
7.	953619114014	DINESH R	D. Dinesh
8.	953619114022	KARTHIK MAHADEV	K. Karthik
9.	953619114023	A.KRISHNA KUMAR	A. Krishna Kumar
10.	953619114024	MAADHESH M	M. Maadhesh
11.	953619114028	P.R.VIJAY JAGANATH	P. R. Vijay Jaganath
12.	953619114032	K.RAMKUMAR	K. Ramkumar
13.	953619114034	SARAVANAKUMAR S	S. Saravanakumar
14.	953619114039	C.THANGA DINESH	C. Thanga Dinesh
15.	953619114040	T.K VIJAYA AKASH	T. K. Vijaya Akash


Class Advisor


Faculty Coordinator


HoD/Mechanical

OBJECTIVES:

At the end of the course, the student is expected to:

- Perform energy audit and identify the opportunities for energy savings
- Understand the various solar technologies and its applications
- Learn other alternate sources of energy and its advancements

UNIT I	ENERGY CONSERVATION AND ENERGY AUDITING	8
Energy Scenario - India and World, Energy Consumption - Environmental Impacts, Energy Conservation, Energy Auditing - Industries and Automobiles, Bureau of Energy Efficiency.		
UNIT II	SOLAR TECHNOLOGIES AND INSTRUMENTATION	14
Solar Energy – Nomenclature, Measuring Instruments, Applications; Ray Tracing; Solar PV Technologies – Efficiency calculation, Waste Management and recycling.		
UNIT III	GREEN BUILDINGS AND RECENT ADVANCEMENTS	8
Green Buildings; Green energy technologies – Other sources of energy, Research focus and recent advancements.		
		TOTAL: 30 PERIODS

OUTCOMES:

Upon the completion of this course the students will be able to

1. Conduct energy audit.
2. Demonstrate possible energy saving opportunities in industries and automobiles.
3. Illustrate solar measuring instruments.
4. Compare the performance of solar applications by simulation and experimentation.
5. Describe recent advancements in green energy technologies

Assessment Method (Weightage 100 %)			
In- Semester	Assessment I	Assessment II	Total
	MCQ / Short Q & A	MCQ / Short Q & A	
	50 %	50 %	
			100 %

REFERENCES:

1. Energy Manager Training Manual (4 Volumes) available at www.energymanagertraining.com, a website administered by Bureau of Energy Efficiency (BEE), a statutory body under Ministry of Power, Government of India, 2004.
2. Sukhatme. S.P., "Solar Energy", Tata McGraw Hill Publishing Company Ltd., New Delhi, 1997.
3. Tiwari. G.N., Solar Energy – "Fundamentals Design, Modelling & Applications", Narosa Publishing House, New Delhi, 2002.


Course Designer & Coordinator


HOD/ Mechanical


Vice-Principal


Principal

RAMCO INSITIUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

VALUE ADDED COURSE TIME TABLE

(Academic year 2021-2022)

Title of the Course : MEVAC1902- Green Energy Technologies and Management

Year / Branch / Semester : III / Mechanical / V

Course Handling Faculty

Industrial Experts:

1. Mr.R.Nesamani, Joint Director, Petroleum Conservation and Research Association
2. Mr.M.Vivekanandan (Certified Energy Auditor), CEO, TryCAE, Trichy

Eminent Academicians from other institutions:

1. Dr. M.Premalatha, Professor, Department of Energy and Environment, NITT
2. Dr.V.Saravanan, Professor, Department of Electrical and Electronics Engineering, Thiagarajar College of Engineering
3. Mr.R.Vignesh Kumar, Assistant Professor, Department of Mechanical Engineering, National Engineering College, Kovilpatti

Internal Faculty Members :

1. Dr.V.Sivakumar, Associate Professor / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam
2. Mr.M.Sivagaminathan alias Balaji, Assistant Professor (Sr.Gr.) / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam
3. Mr.M.Ashok Kumar, Assistant Professor (Sr.Gr.) / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam
4. Mr.R.Arun Kumar, Assistant Professor / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam

Evaluation Team Members: 1. Head of the Department
2. One senior faculty member handling the course
3. One senior faculty member nominated by the Head of the Institution

Session	Date/Day	Topics Covered	Theory (Periods)	Practical (Periods)
I	21.08.2021 (Saturday)	Energy Scenario of India and World, Energy Consumption and Environmental Impacts	3	-
II	22.08.2021 (Sunday)	Energy Conservation, Energy Auditing in Industries and Automobiles	3	-

22/8/21

Session	Date/Day	Topics Covered	Theory (Periods)	Practical (Periods)
III	28.08.2021 (Saturday)	Bureau of Energy Efficiency, Field visit on Energy Auditing in Electrical and Mechanical Utilities	1	2
IV	29.08.2021 (Sunday)	Nomenclature of Solar Energy and Measuring Instruments	3	-
V	04.09.2021 (Saturday)	Solar Energy applications and field exposure on solar thermal applications	1	2
VI	05.09.2021 (Sunday)	A Hands on practice – Solar instruments and experimentation on solar thermal applications	-	3
Assessment I	11.09.2021 (Saturday)	Assessment will be conducted either as MCQ / descriptive type		
VII	12.09.2021 (Sunday)	Introduction and A Hands on practice – Ray Tracing Software	1	2
VIII	18.09.2021 (Saturday)	Introduction to Solar PV Technologies and practical exposure to efficiency calculation on different PV panels	2	1
IX	19.09.2021 (Sunday)	Waste Management and recycling of PV Cells	3	-
X	25.09.2021 (Saturday)	Green Buildings, Field visit on Eco-friendly campus at Ramco Institute of Technology	1	2
XI	26.09.2021 (Sunday)	Other green energy technologies	3	-
XII	02.10.2021 (Saturday)	Research focus and recent advancements in Green Energy Technologies	3	-
Assessment II	03.10.2021 (Sunday)	Assessment will be conducted either as MCQ / descriptive type		
Number of Periods for Theory and Practical Session (Practical session will be consider as 50% of the total periods)			24	12
Total number of periods (2 credit course)			30	


Course Designer & Coordinator


HoD/Mechanical


Vice Principal


Principal



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF MECHANICAL ENGINEERING

Academic year 2021-22 (Odd Semester)

Anna University approved Value Added Course

Willingness Details

Title of the Course : MVA006 Green Energy Technologies and Management

Year / Branch / Semester: III / Mechanical / V

S. No.	Register Number	Name of the Students	Signature
1.	953619114001	ANITH RAJASEELAN	
2.	953619114008	BALAMURUGAN M	
3.	953619114015	DINESHKUMAR. G	
4.	953619114016	ELANGO VAN. K	
5.	953619114017	M GOWTHAM	
6.	953619114019	HARI HARAN VP	
7.	953619114020	JEHAVEERAPANDIYAN	
8.	953619114027	MUTHURAJ M	
9.	953619114030	RAHUL S	
10.	953619114033	K.S.B SANKARA PRASANNA	
11.	953619114036	SHYAMSUNDAR R	
12.	953619114037	SOUMYA SOUVIK KHUNTIA	
13.	953619114041	M.VIJAY THIRUPATHI	
14.	953619114302	A.KANAGARAJ	
15.	953619114303	M.KIRUBAKARAN	
16.	953619114306	MURALIKRISHNAN A	
17.	953619114307	K.RAJASOKANTHAN	

Class Advisor

Faculty Coordinator

HoD/Mechanical

OBJECTIVES:

- To gain knowledge in surface engineering techniques and its applications.
- To classify the different surface coating technologies.
- To enhance skills in understanding the surface characterization techniques and testing methods.

I MODIFICATION BASED SURFACE ENGINEERING METHODS

8

Introduction to Surface Engineering – Classification of Surface Engineering - Modification based surface engineering Methods – Definition and concepts - Annealing (Semiconductor stress relief) – Surface hardening (flame) – Induction Hardening – Shot peening – Diffusion coating – Plasma Ion Implantation – Applications.

II REMOVAL BASED SURFACE ENGINEERING METHODS

7

Definition and Concepts - Electro chemical polishing – Electro discharge machining – Plasma cleaning/etching – Focused ion beam etching – Applications.

III ADDITION BASED SURFACE ENGINEERING METHODS

8

Definition and concepts - Galvanizing – Hot dip coating (Al, Zn) – Diffusion bonding – CVD – PVD – Plasma Spray deposition – Thermal spray deposition – High velocity spray (HVOF / Cold) – Sputtering – Electro deposition/plating – Electroless deposition/ plating – Applications.

IV SURFACE CHARACTERIZATION AND TESTING

7

Scanning Electron microscope surface characterisation – X Ray diffraction - Erosive wear test - Mechanical wear test – Corrosion test.

TOTAL: 30 PERIODS**OUTCOMES:**

Upon the completion of this course, the students will be able to

1. Explain the principles of different coating deposition methods.
2. Differentiate the various surface engineering methods with its significance.
3. Analyse the surface morphology of coated materials with surface characterization techniques.

Assessment Method (Weightage 100 %)			
In-Semester	Assessment I	Assessment II	
	Objective Type Questions (100 Marks)	Objective Type Questions (100 Marks)	Total
	50 %	50 %	100 %

REFERENCES:

1. ASM Handbook: Materials Characterization, ASM International, 2008
2. Sudarshan T S, 'Surface modification technologies - An Engineer's guide', Marcel Dekker, Newyork, 1989.
3. Varghese C.D, 'Electroplating and Other Surface Treatments - A Practical Guide', TMH, 1993.
4. P. Martin, Introduction to Surface Engineering & Functionally Engineered Materials, Wiley, 2011.
5. A.W.Batchelor, N.L. Loh & M.Chandrasekaran, Materials Degradation & Its Control by Surface Engineering, Imperial College Press, 2011.

Kathy S
27/7/21
Course Designer & Coordinator

S. J. K. S.
27/7/21
HOD/ Mechanical

S. J. K. S.
27/7/21
Vice-Principal

M. S. S.
27/7/21
Principal

RAMCO INSTITUTE OF TECHNOLOGY, RAJAPALAYAM**DEPARTMENT OF MECHANICAL ENGINEERING****VALUE ADDED COURSE TIME TABLE****(Academic year 2021-2022)**

Title of the Course : MEVAC1905- Surface Coating Technology

Year / Branch / Semester: III / Mechanical / V

Course Handling Faculty:**Industrial Experts:**

1. Mr.K.R.Rajan, Chief Operating Officer, GowriHouse Metal Works LLP, Rajapalayam.
2. Mr.R.Sakthivel, Manager, GowriHouse Metal Works LLP, Rajapalayam.

Eminent Academicians from other institutions:

1. Dr. J.T. Winowlin Japes, Professor & Dean/ SAME, Kalasalingam University, Krishnan Kovil.
2. Dr. N.Rajini, Professor & Head, Kalasalingam University, Krishnan Kovil.

Internal Faculty Members:

1. Mr.S.Maharajan, Assistant Professor /Mechanical, Ramco Institute of Technology, Rajapalayam
2. Mr.C.Gururaj, Assistant Professor(Sr.Gr.) /Mechanical, Ramco Institute of Technology, Rajapalayam
3. Mr.P.Pavithran, Assistant Professor /Mechanical, Ramco Institute of Technology, Rajapalayam.

Evaluation Team Members: 1. Head of the Department

2. One senior faculty member handling the course

3. One senior faculty member nominated by the Head of the Institution

Session	Date/Day	Topics Covered	Theory Periods	Practical Periods
I	21.08.2021 (Saturday)	Modification Based Surface Engineering Methods. Introduction to Surface Engineering – Classification of Surface Engineering	3	-
II	22.08.2021 (Sunday)	Modification based surface engineering Methods – Definition and concepts - Annealing (Semiconductor stress relief) – Surface hardening (flame)	3	-
III	28.08.2021 (Saturday)	Induction Hardening – Shot peening – Diffusion coating – Plasma Ion Implantation – Applications.	3	-
IV	29.08.2021 (Sunday)	Removal Based Surface Engineering Methods Definition and Concepts - Electro chemical polishing - Electro discharge machining.	3	-

Session	Date/Day	Topics Covered	Theory Periods	Practical Periods
V	04.09.2021 (Saturday)	Plasma cleaning/etching Focused ion beam etching – Applications.	3	-
VI	05.09.2021 (Sunday)	Addition Based surface engineering Methods Definition and concepts - Galvanizing – Hot dip coating (Al, Zn) – diffusion bonding – CVD – PVD – Plasma Spray deposition	3	-
Assessment I	11.09.2021 (Saturday)	Assessment Test I will be conducted for the topics covered from session I to VI	-	-
VII	12.09.2021 (Sunday)	Thermal spray deposition – High velocity spray (HVOF / Cold) - Sputtering – Electro deposition/plating – Electroless deposition/ plating – Applications.	3	-
VIII	18.09.2021 (Saturday)	Practical Demo on Plasma Spray deposition and HVOF spray deposition technique at Spraymet Surface Technologies PVT Ltd, Bangalore.	-	3
IX	19.09.2021 (Sunday)	Surface characterization and Testing Scanning Electron microscope surface characterization – X Ray diffraction	3	-
X	25.09.2021 (Saturday)	Practical Demo on Zinc coating of Screw fasteners at Gowri House Metal Works, Rajapalayam.	-	3
XI	26.09.2021 (Sunday)	Erosive wear test - Mechanical wear test – corrosion test. Practical : Analyze the Surface morphology of coated samples using microscope in Ramco Institute of Technology, Rajapalayam	1	1
XII	02.10.2021 (Saturday)	Practical demo on surface characterization technique at Kalasalingam University.	-	3
Assessment II	03.10.2021 (Sunday)	Assessment Test II will be conducted for the topics covered from session VII to XII.	-	-
Number of Periods for Theory and Practical Sessions (Practical session will be considered as 50% of the total periods)			25	10
Total number of Periods (2 Credit Course)			30	

K. K. S. 30/7/21
Course Designer & Coordinator

S. J. S. 30/7/21
HOD/Mechanical

S. J. S. 30/7/21
Vice-principal

M. S. 30/7/21
Principal



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF MECHANICAL ENGINEERING

Academic year 2021-22 (Odd Semester)

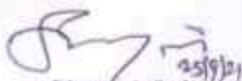
Anna University approved Value Added Course

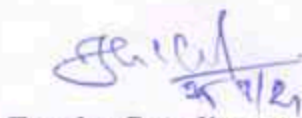
Willingness Details

Title of the Course : MVA009 Surface Coating Technology

Year / Branch / Semester: III / Mechanical / V

S. No.	Register Number	Name of the Students	Signature
1.	953619114002	ARASU T	T. Arasu
2.	953619114004	ARUL PRAKASH K *	K. Arul Prakash
3.	953619114007	ATHITH M	M. Athith
4.	953619114010	BHARATH K	K. Bharath
5.	953619114012	BIJU DANIE S	S. Biju Danie
6.	953619114021	KABIL BHARATH RAM	R. Kabil Bharath
7.	953619114025	MANOJKUMAR K	K. Manoj Kumar
8.	953619114026	MUTHU MARIAPPAN M	M. Muthu Mariappan
9.	953619114029	RAGHUL R	R. Raghul
10.	953619114031	RAMKUMAR.B	B. Ram Kumar
11.	953619114035	SHIRAJ GOKUL N	N. Shiraj Gokul
12.	953619114038	SYEDTHOWFEEK S	S. Syedthowfeek
13.	953619114301	AJITHSANKARRAJ P	P. Ajithsankar
14.	953619114304	MANOJ DHIWAKAR T	T. Manoj Dhiwakar
15.	953619114305	S.MOHAMED ARSATH	S. Mohamed Arsath


Class Advisor


Faculty Coordinator


HoD/Mechanical